

COCHRAN THEOREM

STATEMENT

Let Y is a vector of multi-variate normally distributed and $Y'AY$ is a quadratic form which is partitioned or r quadratic form as follows:

$$Y'AY = Y'A_1Y + Y'A_2Y + \dots + Y'A_rY$$

Then if $\text{Rank}(A) = \sum \text{Rank}(A_i)$

Then each $Y'A_iY \sim \chi^2_r(A_i)$ and all A_i are mutually independent.

TESTING RCBD MODEL

$$Y = XQ + \varepsilon \Rightarrow \varepsilon = Y - XQ$$

We have

$$SSE = \varepsilon'\varepsilon$$

$$= (Y - XQ)'(Y - XQ)$$

$$= (Y' - X'Q')(Y - XQ)$$

$$= Y'Y - QY'X - Q'X'Y + Q'X'XQ$$

$$= Y'Y - 2Q'X'Y + Q'X'XQ$$

After least squares methods replacing Q by $\hat{Q}$

$$SEE = Y'Y - 2\hat{\theta}'X'Y + \hat{\theta}'X'X\hat{\theta}$$

putting value of $\hat{\theta} = (X'X)^{-1}X'Y$

$$= Y'Y - 2\hat{\theta}'X'Y + \hat{\theta}'\underbrace{X'X(X'X)^{-1}}_I X'Y$$

$$= Y'Y - 2\hat{\theta}'X'Y + \hat{\theta}'X'Y$$

$$= Y'Y - \hat{\theta}'X'Y$$

↓

↓

Total SS

SS due to model.

As

$$\therefore \hat{\theta} = \begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} \Rightarrow \hat{\theta}' = (\hat{\alpha}' \hat{\beta}')$$

$$\therefore X'Y = \begin{pmatrix} G \\ T \\ B \end{pmatrix}$$

So,

$$SSE = Y'Y - [\hat{\alpha}' \hat{\beta}'] \begin{pmatrix} G \\ T \\ B \end{pmatrix}$$

$$SSE = Y'Y - (\hat{\alpha}'G + \hat{\beta}'T + \hat{\beta}'B)$$

$$\therefore \hat{\alpha} = \frac{G}{tr}, \quad \hat{\beta} = \left(\frac{T}{r_2} - \frac{1tG}{tr} \right)$$

$$\therefore \hat{\beta} = \left(\frac{B}{t} - \frac{1r_2G}{tr} \right)$$

$$SSE = Y'Y - \left[\frac{G'G}{tr} + \left(\frac{T}{r} - \frac{1tG}{tr} \right)' \frac{T}{tr} + \left(\frac{B}{t} - \frac{1tG}{tr} \right)' \frac{B}{tr} \right]$$

$$= Y'Y - \left[\frac{G'G}{tr} + \left(\frac{T'T}{r} - \frac{G'1'tT}{tr} \right) + \left(\frac{B'B}{t} - \frac{G'1'rB}{tr} \right) \right]$$

NOW,

$$1t'T = (1 \ 1 \ \dots \ 1) \begin{pmatrix} T_1 \\ T_2 \\ \vdots \\ T_r \end{pmatrix} \quad (t \times r)$$

$$1t'T = T_1 + T_2 + \dots + T_r$$

$$1t'T = \sum_{i=1}^r T_i$$

$$\boxed{1t'T = G}$$

$$1'rB = (1 \ 1 \ \dots \ 1) \begin{pmatrix} B_1 \\ B_2 \\ \vdots \\ B_r \end{pmatrix} \quad (t \times r)$$

$$= B_1 + B_2 + \dots + B_r$$

$$= \sum_{j=1}^r B_j$$

$$\boxed{1'rB = G}$$

So,

$$SSE = \left(\frac{Y'Y - G'G}{t_2} \right) - \left(\frac{T'T - G'G}{n} \right) - \left(\frac{B'B - G'G}{t} \right)$$

$SSE = \text{Total SS} - \text{Treatment SS} - \text{Block SS}$

$\text{Total SS} = \text{Treatment SS} + \text{Block SS} + \text{Error SS}$

$$\left(\frac{Y'Y - G'G}{t_2} \right) = \left(\frac{T'T - G'G}{n} \right) + \left(\frac{B'B - G'G}{t} \right) + \left(\frac{Y'Y - T'T - B'B + G'G}{nt} \right)$$

As,

$$G = X_1'Y \Rightarrow G' = Y'X_1$$

$$T = X_2'Y \Rightarrow T' = Y'X_2$$

$$B = X_3'Y \Rightarrow B' = Y'X_3$$

So,

$$\left(\frac{Y'Y - \frac{Y'X_1X_1'Y}{nt}}{nt} \right) = \left(\frac{\frac{Y'X_2X_2'Y}{n} - \frac{Y'X_1X_1'Y}{nt}}{nt} \right) + \left(\frac{\frac{Y'X_3X_3'Y}{t} - \frac{Y'X_1X_1'Y}{nt}}{nt} \right) + \left(\frac{Y'Y - \frac{Y'X_2X_2'Y}{n} - \frac{Y'X_3X_3'Y}{t} + \frac{Y'X_1X_1'Y}{nt}}{nt} \right)$$

$$\Rightarrow Y' \left(I_{2t} - \frac{X_1 X_1'}{2t} \right) Y = Y' \left(\frac{X_2 X_2'}{2} - \frac{X_1 X_2'}{2t} \right) Y + Y' \left(\frac{X_3 X_3'}{t} - \frac{X_1 X_3'}{2t} \right) Y$$

$$+ Y' \left(I_{2t} - \frac{X_2 X_2'}{2} - \frac{X_3 X_3'}{t} + \frac{X_1 X_1'}{2t} \right) Y$$

Let

$$A = I_{2t} - \frac{X_1 X_1'}{2t}$$

$$A_1 = \frac{X_2 X_2'}{2} - \frac{X_1 X_1'}{2t}$$

$$A_2 = \frac{X_3 X_3'}{t} - \frac{X_1 X_1'}{2t}$$

$$A_3 = I_{2t} - \frac{X_2 X_2'}{2} - \frac{X_3 X_3'}{t} + \frac{X_1 X_1'}{2t}$$

Then

$$Y' A Y = Y' A_1 Y + Y' A_2 Y + Y' A_3 Y$$

By Cochran theorem

$$\text{Rank}(A) = \text{Rank}(A_1) + \text{Rank}(A_2) + \text{Rank}(A_3) \quad \text{--- (1)}$$

Now $\text{Rank}(A) = \text{tr}(A)$ if A is independent
ie $A^2 = A$

$$A^2 = (A)(A)$$

$$= \left(I_{2t} - \frac{X_1 X_1'}{2t} \right) \left(I_{2t} - \frac{X_1 X_1'}{2t} \right)$$

$$= I_{rt} - \frac{X_1 X_1'}{rt} - \frac{X_1 X_1'}{rt} + \frac{X_1 X_1' X_1 X_1'}{(rt)^2}$$

$$= I_{rt} - \frac{2X_1 X_1'}{rt} + \frac{X_1 rt X_1'}{(rt)^2} \quad \because X_1' X_1 = rt$$

$$= I_{rt} - \frac{2X_1 X_1'}{rt} + \frac{X_1 X_1'}{rt}$$

$$= I_{rt} - \frac{X_1 X_1'}{rt} = A$$

So A is idempotent and hence

$$\text{Rank}(A) = \text{tr}(A)$$

$$= \text{tr}\left(I_{rt} - \frac{X_1 X_1'}{rt}\right)$$

$$= \text{tr}(I_{rt}) - \text{tr}\left(\frac{X_1 X_1'}{rt}\right)$$

$$\therefore \text{tr}(I_{rt}) = rt$$

$$\therefore \text{tr}(X_1 X_1') = \text{tr}(X_1' X_1)$$

$$\therefore X_1' X_1 = rt$$

$$= rt - \text{tr}\left(\frac{X_1' X_1}{rt}\right)$$

$$= rt - \text{tr}\left(\frac{rt}{rt}\right)$$

$$\boxed{\text{Rank}(A) = rt - 1} \quad \text{--- (2)}$$

Now,

$$\text{Rank}(A_1) = \text{tr}(A_1) \text{ of } A_1^2 = A_1$$

$$A_1 = \begin{pmatrix} \frac{x_2 x_2'}{r_2} & -\frac{x_1 x_1'}{r_2 t} \\ \frac{x_2 x_2'}{r_2} & -\frac{x_1 x_1'}{r_2 t} \end{pmatrix}$$

$$A_1^2 = A_1 A_1$$

$$= \begin{pmatrix} \frac{x_2 x_2'}{r_2} & -\frac{x_1 x_1'}{r_2 t} \\ \frac{x_2 x_2'}{r_2} & -\frac{x_1 x_1'}{r_2 t} \end{pmatrix} \begin{pmatrix} \frac{x_2 x_2'}{r_2} & -\frac{x_1 x_1'}{r_2 t} \\ \frac{x_2 x_2'}{r_2} & -\frac{x_1 x_1'}{r_2 t} \end{pmatrix}$$

$$= \frac{x_2 x_2' x_2 x_2'}{r_2^2} - \frac{x_2 x_2' x_1 x_1'}{r_2^2 t} - \frac{x_1 x_1' x_2 x_2'}{r_2^2 t} + \frac{x_1 x_1' x_1 x_1'}{(r_2 t)^2}$$

$$A_1^2 = \frac{x_2 r_2 t x_2'}{r_2^2} - \frac{x_2 r_2 t x_1'}{r_2^2 t} - \frac{x_1 r_2 t x_2'}{r_2^2 t} + \frac{x_1 r_2 t x_1'}{(r_2 t)^2}$$

$$\therefore x_2' x_2 = r_2 t$$

$$\therefore x_2' x_1 = r_2 t$$

$$\therefore x_1' x_2 = r_2 t$$

$$\therefore x_1' x_1 = r_2 t$$

$$x_2 t = (t x_2) (t x_1)$$

$$x_2 t = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} (t \times 1)$$

$$x_2 t = (t x_2) (t x_1)$$

$$= t x_1$$

$$= x_1$$

$$1' t x_2' = (1 x t) (t x t_2)$$

$$= t x t_2$$

$$= x_1'$$

$$A_1^2 = \frac{x_2 x_2'}{n} - \frac{\frac{1}{n} x_1 x_1'}{n't} - \frac{\frac{1}{n} x_1 x_1'}{n't} + \frac{x_1 x_1'}{n't}$$

$$A_1^2 = \frac{x_2 x_2'}{n} - \frac{x_1 x_1'}{n't} = A_1$$

So,

$$\text{Rank}(A_1) = \text{Tr}(A_1)$$

$$= \text{tr} \left[\frac{x_2 x_2'}{n} - \frac{x_1 x_1'}{n't} \right]$$

$$= \text{tr} \left(\frac{x_2 x_2'}{n} \right) - \left(\text{tr} \left(\frac{x_1 x_1'}{n't} \right) \right)$$

$$= \text{tr} \left(\frac{x_2' x_2}{n} \right) - \text{tr} \left(\frac{x_1' x_1}{n't} \right)$$

$$= \text{tr} \left(\frac{n't}{n} \right) - \text{tr} \left(\frac{n't}{n't} \right)$$

$$= \text{tr}(I_t) - \text{tr}(1)$$

$$\boxed{\text{Rank}(A_1) = t - 1}$$

Now,

$$\text{Rank}(A_2) = \text{tr}(A_2) \text{ if } A_2^2 = A_2$$

$$A_2 = \begin{pmatrix} \frac{x_3 x_3'}{t} & -\frac{x_1 x_1'}{n't} \\ & \end{pmatrix}$$

$$A_2^2 = A_2 A_2$$

$$= \begin{pmatrix} \frac{x_3 x_3'}{t} & -\frac{x_1 x_1'}{n't} \\ & \end{pmatrix} \begin{pmatrix} \frac{x_3 x_3'}{t} & -\frac{x_1 x_1'}{n't} \\ & \end{pmatrix}$$

$$A_2^2 = \frac{X_3 X_3' X_3 X_3'}{t^2} - \frac{X_3 X_3' X_1 X_1'}{2t^2} - \frac{X_1 X_1' X_3 X_3'}{2t^2} + \frac{X_1 X_1' X_1 X_1'}{(2t)^2}$$

$$A_2^2 = \frac{X_3 t I_2 X_3'}{t^2} - \frac{X_3 t I_2 X_1'}{2t^2} - \frac{X_1 t I_2' X_3'}{2t^2} + \frac{X_1 2t X_1'}{(2t)^2}$$

$$\therefore X_1' X_1 = t I_2$$

$$\therefore X_1' X_3 = t I_2'$$

$$\therefore X_3' X_1 = t I_2$$

$$\therefore X_3' X_3 = t I_2$$

$$A_2^2 = \frac{X_3 t I_2 X_3'}{t^2} - \frac{X_3 t I_2 X_1'}{2t^2} - \frac{X_1 t I_2' X_3'}{2t^2} + \frac{X_1 2t X_1'}{(2t)^2}$$

$$\therefore X_3 t I_2 = (t I_2 X_3')(2 X_1)$$

$$= t I_2 X_1 = X_1$$

$$\therefore I_2' X_3' = (I_2 X_3')(2 X_1 t I_2)$$

$$= (I_2 X_3 t I_2) = X_1'$$

$$A_2^2 = \frac{X_3 X_3'}{t} - \frac{X_1 X_1'}{2t} - \frac{X_1 X_1'}{2t} + \frac{X_1 X_1'}{2t}$$

$$= \frac{X_3 X_3'}{t} - \frac{X_1 X_1'}{2t}$$

$$= A_2$$

So,

$$\text{Rank}(A_2) = \text{tr}(A_2)$$

$$= \text{tr} \left(\frac{X_3 X_3'}{t} - \frac{X_1 X_1'}{rt} \right)$$

$$= \text{tr} \left(\frac{X_3 X_3'}{t} \right) - \text{tr} \left(\frac{X_1 X_1'}{rt} \right)$$

$$= \text{tr} \left(\frac{X_3' X_3}{t} \right) - \text{tr} \left(\frac{X_1' X_1}{rt} \right)$$

$$= \text{tr} \left(\frac{k I_2}{k} \right) - \text{tr} \left(\frac{rt}{rt} \right) \because X_3' X_3 = t I_2$$

$$= \text{tr}(I_2) - \text{tr}(1)$$

$$\boxed{\text{Rank}(A_2) = r - 1} \quad \text{--- (4)}$$

$$\text{Rank}(A_3) = \text{tr}(A_3) \text{ if } A_3^2 = A_3$$

$$A_3 = \begin{bmatrix} I_2 t - \frac{X_2 X_2'}{r} - \frac{X_3 X_3'}{t} + \frac{X_1 X_1'}{rt} \end{bmatrix}$$

$$A_3^2 = (A_3)(A_3)$$

$$= \left(I_2 t - \frac{X_2 X_2'}{r} - \frac{X_3 X_3'}{t} + \frac{X_1 X_1'}{rt} \right) \left(I_2 t - \frac{X_2 X_2'}{r} - \frac{X_3 X_3'}{t} + \frac{X_1 X_1'}{rt} \right)$$

$$= I_2 t - \frac{X_2 X_2'}{r} - \frac{X_3 X_3'}{t} + \frac{X_1 X_1'}{rt} - \frac{X_2 X_2'}{r} + \frac{X_2 X_2' X_2 X_2'}{r^2}$$

$$+ \frac{X_2 X_2' X_3 X_3'}{s^2 t} - \frac{X_2 X_2' X_1 X_1'}{s^2 t} - \frac{X_3 X_3'}{t} + \frac{X_2 X_3' X_2 X_2'}{t s}$$

$$+ \frac{X_3 X_3' X_3 X_3'}{t^2} - \frac{X_3 X_3' X_1 X_1'}{s^2 t^2} + \frac{X_1 X_1'}{s^2 t} - \frac{X_1 X_1' X_2 X_2'}{s^2 t}$$

$$- \frac{X_1 X_1' X_3 X_3'}{s^2 t^2} + \frac{X_1 X_1' X_1 X_1'}{(s^2 t)^2}$$

$$A_3^2 = I s^2 t - \frac{X_2 X_2'}{s^2 t} - \frac{X_3 X_3'}{t} + \frac{X_1 X_1'}{s^2 t} - \frac{X_2 X_2'}{s^2 t} + \frac{X_2 s^2 t X_2'}{s^2 t^2}$$

$$+ \frac{X_2 t I s^2 X_3'}{s^2 t} - \frac{X_2 s^2 t X_1'}{s^2 t^2} - \frac{X_3 X_3'}{t} + \frac{X_3 t s^2 t X_2'}{s^2 t^2}$$

$$+ \frac{X_3 t I s^2 X_2'}{t^2} - \frac{X_3 t s^2 t X_1'}{s^2 t^2} + \frac{X_1 X_1'}{s^2 t} - \frac{X_1 s^2 t X_2'}{s^2 t^2}$$

$$- \frac{X_1 t s^2 t X_3'}{s^2 t^2} + \frac{X_1 s^2 t X_1'}{s^2 t^2}$$

For Simplification

$$\begin{array}{ll} \text{order of } n_1 & s^2 t X_1 \\ n_2 & s^2 t X_2 \\ n_3 & s^2 t X_3 \end{array}$$

$$\begin{array}{l} X_1' X_1 = s^2 t, X_1' X_2 = s^2 t I, X_1' X_3 = t I s^2 \\ X_2' X_1 = s^2 t, X_2' X_2 = s^2 I t, X_2' X_3 = t I s^2 \\ X_3' X_1 = t I s^2, X_3' X_2 = t I s^2, X_3' X_3 = t I s^2 \end{array}$$

$$① X_2 1t \Rightarrow (2t \times t)(t \times 1) \Rightarrow (2t \times 1) \Rightarrow X_1$$

$$② 1'2X_3' \Rightarrow (1 \times 2)(2 \times 2t) \Rightarrow (1 \times 2t) \Rightarrow X_1' \quad \text{--- ⑧}$$

$$③ X_2 1t \Rightarrow (2t \times t)(t \times 1) \Rightarrow (2t \times 1) \Rightarrow X_1$$

$$⑤ 1tX_2' \Rightarrow (1 \times t)(t \times 2t) \Rightarrow (1 \times 2t) \Rightarrow X_1' \quad \text{--- ⑦}$$

$$④ X_3 12 \Rightarrow (2t \times 2)(2 \times 1) \Rightarrow (2t \times 1) \Rightarrow X_1 \quad \text{--- ⑥}$$

So,

$$A_3^2 = I_2 - \frac{X_2 X_2'}{2} - \frac{X_3 X_3'}{t} + \frac{X_1 X_1'}{2t} - \frac{X_2 X_3'}{2} + \frac{X_2 X_1'}{2}$$

$$+ \frac{X_1 X_2'}{2t} - \frac{X_1 X_3'}{2t} - \frac{X_3 X_1'}{t} + \frac{X_1 X_1'}{2t}$$

$$+ \frac{X_3 X_2'}{t} - \frac{X_1 X_2'}{2t} + \frac{X_1 X_3'}{2t} - \frac{X_1 X_1'}{2t} - \frac{X_1 X_1'}{2t}$$

$$+ \frac{X_1 X_1'}{2t}$$

$$= A_3$$

$$\text{So Rank}(A_3) = \text{tr}(A_3)$$

$$= \text{tr} \left(I_2 - \frac{X_2 X_2'}{2} - \frac{X_3 X_3'}{t} + \frac{X_1 X_1'}{2t} \right)$$

$$= \text{tr}(I_2) - \text{tr} \left(\frac{X_2 X_2'}{2} \right) - \text{tr} \left(\frac{X_3 X_3'}{t} \right) + \text{tr} \left(\frac{X_1 X_1'}{2t} \right)$$

$$= \text{tr}(I_2) - \text{tr} \left(\frac{X_2' X_2}{2} \right) - \text{tr} \left(\frac{X_3' X_3}{t} \right) + \text{tr} \left(\frac{X_1' X_1}{2t} \right)$$

$$= r_1 t - t_2 \left(\frac{r_1 I_1 t}{r_1} \right) - t_2 \left(\frac{r_2 I_2}{r_2} \right) + t_2 \left(\frac{r_2 t}{r_2} \right)$$

$$= r_1 t - t_2 (I_1 t) - t_2 (I_2) + t_2 (1)$$

$$= r_1 t - t - r_2 + 1$$

$$= t(r_1 - 1) - 1(r_2 - 1)$$

$$\text{Rank}(A_3) = (t-1)(r_2-1) \quad \text{--- (5)}$$

Using (2), (3), (4) and (5) in (1)

$$r_1 t - 1 = (t-1) + (r_2-1) + (r_1 t - t - r_2 + 1)$$

$$= t - 1 + r_2 - 1 + r_1 t - t - r_2 + 1$$

$$r_1 t - 1 = r_1 t - 1$$

Hence

$$\text{Rank}(A) = \text{Rank}(A_1) + \text{Rank}(A_2) + \text{Rank}(A_3)$$

Since

$$Y \sim N(X\beta, \sigma^2 I)$$

So,

$$Y' A_1 Y \sim \chi^2(t-1)$$

$$Y' A_2 Y \sim \chi^2(r_2-1)$$

$$Y' A_3 Y \sim \chi^2(t-1)(r_2-1)$$

By Cochran Theorem

Now,

$$F_1 = \bar{F}[(t-1), (t-1)(r_2-1)]$$

$$= \frac{Y' A_1 Y / (t-1)}{Y' A_3 Y / (t-1)(r_2-1)}$$

$$= \frac{Y' A_1 Y / (t-1)}{Y' A_3 Y / (t-1)(r_2-1)}$$

$$= \frac{MST}{MSE}$$

$$MSE$$

$$F_2 = F((2-1), (t-1)(2-1))$$

$$= \frac{Y'A_2Y / (2-1)}{Y'A_3Y / (t-1)(2-1)}$$

$$= \frac{SSB}{MSE}$$

$$= \frac{SSB}{MSE}$$

$$MSE$$

ANOVA TABLE

S.V	df	SS	MS	F-ratio
Treat- ment	(t-1)	$T'T - \frac{G'G}{t}$ = SST	$SST / (t-1)$ = MST	$F_1 = \frac{MST}{MSE}$
Block	(2-1)	$B'B - \frac{G'G}{t}$ = SSB	$SSB / (2-1)$ = MSB	$F_2 = \frac{MSB}{MSE}$
Error	(t-1)(2-1)	$Y'Y - T'T - B'B + \frac{G'G}{t}$ = SSE	$SSE / (t-1)(2-1)$ = MSE	---
Total	2t-1	$Y'Y - \frac{G'G}{t}$ = SST	---	---